INTRODUCTION

Cooling systems are a critical component of telecom central offices, where stable environmental conditions are required for the reliable and continuous operation of sensitive electronic equipment. In such environments, cooling is not a comfort-driven service but an operational necessity, and its energy use forms a substantial share of total site demand. Telecom-specific studies and adjacent data-centre research show that supporting systems, especially cooling, are a major component of infrastructure electricity consumption (Sorrentino *et al.*, 2019; D'Aniello *et al.*, 2018; Ni and Bai, 2017). As a result, cooling performance is not a secondary technical issue, but an important part of site-level energy management.

Modern telecom facilities generate large volumes of monitoring data, including subsystem energy signals, environmental variables, and operational indicators. However, these data are mostly used for monitoring, visualisation, and diagnostics rather than for predictive or supervisory decision-making. At the same time, improving energy efficiency has become increasingly important under cost, sustainability, and reliability constraints. This creates a practical need for analytical approaches that can transform existing telemetry into defensible operational insight without compromising thermal safety.

A further challenge lies in the nature of the available data. In real monitored infrastructure, cooling demand is often not available as a ready-made hourly variable. Instead, the analytical target may need to be reconstructed from fragmented cumulative subsystem measurements, incomplete contextual signals, and uneven data quality. The reviewed literature shows that telecom studies are comparatively well developed in monitoring, KPI design, and diagnosis, but less developed in the full chain from target engineering to short-horizon cooling-demand forecasting, expected-versus-actual benchmarking, and forecast-informed supervisory control under sensing constraints (Qaisar and Zhao, 2022; Malafronte, Sorrentino and Trifirò, 2021; Mea *et al.*, 2017; Sorrentino *et al.*, 2016). The present thesis addresses this gap by combining these strands within one real telecom central office case.

The aim of the thesis is to develop predictive models for cooling energy use and evaluate intelligent control strategies based on real monitoring data from telecom central office environments. This aim is pursued through hourly target engineering from fragmented telemetry, short-term forecasting, benchmark-based assessment of actual versus expected cooling demand, and simulation of forecast-informed intelligent control strategies.

Accordingly, the thesis is positioned as an applied study of forecast-informed supervisory control rather than as full model predictive control deployment, reinforcement-learning control, or actuator-level closed-loop validation. In this thesis, intelligent control strategies are implemented as forecast-informed strategy simulations evaluated on historical operation rather than as deployed closed-loop controllers. This scope is methodologically consistent with the available telemetry and with the research gap identified in the literature review.

To achieve this aim, the thesis solves the following **research tasks**:

1. To conduct a literature review on predictive modelling, benchmark-based evaluation, and intelligent control strategies for cooling systems.
2. To explore, preprocess, and analyse real operational and environmental data, and to engineer an hourly cooling-demand target and analytical dataset from fragmented site telemetry.
3. To develop and compare short-term forecasting models for cooling energy use.
4. To analyse actual versus expected cooling demand using the selected forecasting model as a benchmark.
5. To design, simulate, and evaluate forecast-informed intelligent control strategies against historical operation.

The research questions of the thesis are:

1. How can real operational and environmental data from a telecom central office be analysed and transformed into a defensible hourly cooling-demand target and analytical dataset?
2. Which forecasting approach provides the most reliable short-term benchmark for expected cooling demand?
3. To what extent can forecast-informed intelligent control strategies, evaluated through historical simulation, identify conservative opportunities to improve cooling energy use?

The object of the research is cooling energy use in telecom central office technical infrastructure.

The subject of the research is data-driven prediction, benchmark-based assessment, and intelligent control strategy simulation for cooling energy use in telecom central office environments.

Commented [AZ1]: Real operational and environmental data can be transformed into a defensible hourly cooling-demand target by screening signals for availability, continuity, meaning and completeness; selecting cooling-related cumulative subsystem meters; aligning them to hourly timestamps; converting cumulative measurements into interval consumption through differencing; clipping negative or implausible steps; aggregating the retained hourly increments into the final target y; and then constructing an analytical dataset with weather, calendar and operational context variables. In the final experiment, this target was engineered from subsystem signals 120, 230, 46, 49 and 78, and the resulting main dataset was shown to be sufficiently representative for the forecasting and benchmarking tasks.

Commented [AZ2]: The most reliable short-term benchmark in the final experiment is Gradient Boosting with lagged target features, selected on the basis of rolling-origin backtesting. Random Forest remained the strongest secondary benchmark and the most useful interpretive model, while naive persistence remained a serious short-horizon baseline because the process is strongly persistence-driven. A limited six-hour-ahead test was retained only as a sensitivity check, so the main answer clearly belongs to the one-hour horizon. The selected final model achieved MAE ≈ 117.4, RMSE ≈ 166.3, CVRMSE ≈ 4.92%, NMAE ≈ 3.48%, and $R^2 \approx 0.985$

Commented [AZ3]: Forecast-informed intelligent control strategies can identify a small, bounded, and context-aware set of conservative intervention opportunities, but they do not support a claim of large reducible cooling waste at the selected site. The final hierarchy shows that simpler strategies are more aggressive, whereas context-aware strategies are more selective and safer. After adding internal_temp_proxy, the final S2 and S3 strategies eliminated high-risk actions, while S3 became more refined rather than simply more restrictive.

The methods used in the thesis are:

1. Documented literature search and structured literature review on predictive modelling and intelligent control strategies in cooling energy management.
2. Preprocessing, exploratory analysis, signal screening, and target engineering of real operational and environmental data using Python-based analytical tools.
3. Development and comparison of short-term forecasting models using time-series and machine-learning techniques selected on the basis of literature and data characteristics.
4. Validation of forecasting models using chronological and rolling-origin evaluation with MAE, RMSE, CVRMSE, NMAE, and R^2.
5. Forecast-based benchmarking of actual versus expected cooling demand.
6. Historical simulation and sensitivity-based evaluation of intelligent control strategies through comparison with baseline operation.

The expected result of the thesis is a defensible analytical framework for telecom cooling analytics that integrates engineered cooling-demand telemetry, short-horizon forecasting, benchmark-based operational assessment, and forecast-informed intelligent control strategy simulation. The framework is intended to support the identification of conservative and context-aware opportunities for improving cooling energy use under real telemetry constraints.

The thesis is structured as follows. Chapter 1 reviews the literature on cooling energy in telecom central offices and related technical infrastructure, operational data quality and target engineering, short-term cooling-demand forecasting, intelligent control strategies, and forecast-based benchmarking. Chapter 2 describes the research design, case-study setting, data sources, signal screening, target engineering procedure, dataset construction, exploratory analysis, forecasting methodology, and benchmarking and supervisory-control methodology. Chapter 3 presents the forecasting results, including model comparison, rolling-origin validation, final model selection, forecast quality, and interpretive feature analysis. Chapter 4 presents the benchmark-based and supervisory-control simulation results, including expected-versus-actual cooling analysis, reduction-potential filtering, comparative strategy results, and sensitivity analysis. Chapter 5 discusses the main findings, answers the research questions, states the contribution of the thesis, and identifies limitations and future research directions. The thesis ends with conclusions, the list of references and sources, and annexes.

1. LITERATURE REVIEW

1.1 Literature search strategy and scope

This literature review was guided by a documented search strategy designed to identify studies relevant to data-driven prediction and analysis of cooling energy use in telecom central office environments. The review focused on five thematic areas that structure the thesis: telecom cooling diagnostics, operational data preprocessing and target engineering, short-term cooling-demand forecasting, benchmark-based evaluation, and intelligent control strategies for cooling systems. Because telecom-specific evidence remains limited, relevant transfer literature from data centres and building HVAC research was also included when it contributed directly to the methodological framing of forecasting, benchmarking, or intelligent control under operational and thermal constraints.

Search terms were organised into concept groups covering domain, system, outcome, method, and operational context. The domain group included *telecom**, "telecommunication central office", "data center", *datacenter*, "server room", and "mission-critical". The system group included *cool**, HVAC, "air condition*", and *chiller**. The outcome group included "energy consumption", "cooling load", "power demand", "energy efficien*", "baseline", and "anomaly detection". The method group included *forecast**, *prediction*, "time series", ARIMA, LSTM, "gradient boosting", "random forest", "model predictive control", MPC, "rule-based", and "reinforcement learning". These groups were combined using Boolean OR within groups and AND between groups. The main databases were Scopus and Google Scholar. Scopus was used for structured filtering and metadata control, while Google Scholar was used to broaden recall and support backward and forward citation chaining. AI-assisted tools were used only to suggest query variants. All inclusion decisions were verified against the original source (Martín-Martín *et al.*, 2018).

Studies were included when they were available in full text, written in English, and published mainly between 2015 and 2025, while older seminal studies were retained only when methodologically necessary. Priority was given to peer-reviewed empirical studies, comparative model-evaluation studies, and review papers that provided essential theoretical framing. Eligible studies addressed cooling energy use, cooling-load forecasting, baseline or anomaly modelling, or intelligent control of cooling and HVAC systems in telecom, data-centre, or closely related technical-infrastructure settings. Studies were excluded if they focused only on heating, lacked direct relevance to cooling-energy analysis, were purely descriptive without analytical or methodological contribution, duplicated a more complete version of the same study or had

inaccessible full text. The review question, keyword design, database selection, and screening logic were defined in advance to improve transparency and reproducibility (Page *et al.*, 2021; Okoli, 2015).

The search process combined database searching with backward and forward citation tracking from core papers. Records were first screened by title and abstract for topical relevance and were then assessed in full text against the inclusion and exclusion criteria. The final set of studies was organised thematically according to the structure of this thesis: telecom diagnostics, target engineering, forecasting, benchmark-oriented interpretation, and intelligent control. This strategy was designed to balance breadth, methodological relevance, and reproducibility for a real-data telecom cooling study.

1.2 Cooling energy in telecom central offices and related technical infrastructure

Telecommunication central offices are highly energy-intensive technical environments in which cooling is an operational necessity rather than a comfort-driven service. In such sites, total electricity use is typically divided into direct consumption associated with telecommunication equipment and indirect consumption associated mainly with supporting systems, especially cooling. Telecom-specific studies indicate that this indirect share can become very large, in some cases approaching half of total grid-supplied energy, which makes cooling performance a critical part of site-level energy management rather than a secondary auxiliary issue (Sorrentino *et al.*, 2019; D'Aniello *et al.*, 2018). This broader importance is consistent with adjacent data-centre literature, where air-conditioning systems account on average for about 38% of facility energy use and are therefore a major target for efficiency assessment and benchmarking (Ni and Bai, 2017).

The telecom literature has approached this problem primarily through monitoring, diagnostics and energy-intelligence frameworks. Sorrentino et al. (2016) developed a model-based approach for monitoring and diagnosis of telecommunication cooling systems, while Mea et al. (2017) extended the same line of work toward real-time monitoring, diagnosis and offline optimisation analysis at room level. D'Aniello et al. (2018) proposed high-level energy performance metrics for telecommunication sites, including indicators explicitly designed to identify anomalous energy absorption and data reliability problems. Malafronte et al. (2021) further showed that meaningful diagnosis can be achieved largely from existing electricity measurements, without major expansion of the sensing infrastructure and reported measurable yearly savings in a high-demand central office after deployment of the proposed diagnostic logic.

A consistent theme in this telecom literature is the use of synthetic, benchmark-oriented indicators to translate complex operational data into interpretable diagnostic measures. D'Aniello et al. (2018) combine reliability- and benchmark-oriented indices to distinguish unreliable measurements from genuinely anomalous site behaviour, while Sorrentino et al. (2019) develop the utilization factor as a climate-independent metric tailored to the energy breakdown of telecommunication central offices. Malafronte et al. (2021) extend this direction by showing that diagnosis-oriented energy intelligence can be built largely from existing electricity measurements, without major expansion of the sensing infrastructure. Together, these studies frame telecom cooling not simply as a load to be monitored, but as an operational process to be assessed against reference behaviour using data-driven diagnostic logic.

This telecom KPI and diagnosis line has also been extended into longer-horizon energy-management support. Sorrentino et al. (2024) develop a multi-annual monitoring tool for telecommunication central offices that classifies sites by efficiency and measurement reliability using KPI-based analysis over multi-year periods. This is useful for the present thesis because it strengthens the argument that telecom cooling analytics are not limited to one-off diagnosis, but belong to a broader energy-intelligence tradition in which benchmark-oriented monitoring, reliability assessment and operational interpretation are tightly connected.

Within the reviewed telecom corpus, monitoring, KPI development and diagnosis appear more developed than short-horizon cooling-demand forecasting from fragmented operational telemetry. Existing studies show clearly why cooling energy should be monitored, benchmarked and diagnostically interpreted in telecommunication sites, but they provide less direct support for the full chain from target reconstruction to short-term forecasting and model-based expected-demand benchmarking. This is the point at which the present thesis is positioned. It builds on the telecom tradition of energy intelligence and benchmark-oriented diagnosis, but extends it toward data-driven forecasting and operational assessment of cooling demand in a real central-office environment.

1.3 Operational data quality, preprocessing and target engineering

A recurring finding in the building-energy literature is that operational data are rarely ready for direct modelling. Fan et al. (2021) describe data preprocessing as an indispensable stage of knowledge discovery from building operational data and show that missing values, outliers, reduction, scaling, transformation and partitioning can all materially affect subsequent analysis. They also stress that preprocessing in real buildings cannot be fully standardised because both data

quality and operating characteristics vary across applications. In real industrial telemetry, preprocessing choices also determine whether later forecasting and benchmarking results reflect genuine operational structure or artefacts of missingness, misalignment and sensor instability.

In practice, however, monitored datasets rarely provide a complete and uniformly reliable representation of all relevant variables. The preprocessing literature therefore supports a pragmatic distinction between signals that can be used directly, signals that require transformation and signals that should be excluded because of instability or insufficient completeness. This is particularly important in operational energy analytics, where the validity of later benchmarking depends on whether the retained variables reflect persistent system behaviour rather than defects in data collection or storage.

The most relevant methodological literature for the present thesis is the literature on sub-metering. Li et al. (2023) argue that sub-metering is a primary requirement for energy-efficiency evaluation because total building energy consumption is too aggregated to reveal the behaviour and savings potential of individual sub-systems. Their study is especially important because it treats sub-item prediction and sub-item evaluation as analytically more informative than total-consumption assessment alone. In this view, itemized energy data are not simply a richer dataset, but a different level of observation that allows operational patterns to be assigned to specific end uses rather than being diluted in a whole-building total.

The difficulty of obtaining a usable cooling target is also reflected in submetering-based HVAC assessment studies. Ji et al. (2017) combine electricity submetering with cooling-load estimation in order to assess main HVAC equipment performance, while Li et al. (2023) distinguish direct and indirect metering relationships when reconstructing sub-item energy behaviour. Together, these studies support the view that cooling demand may need to be inferred from distributed measurements rather than directly observed as a single ready-made variable. This is particularly relevant in monitored technical infrastructure, where subsystem-level cumulative meters may be the only consistent trace of cooling activity. In such cases, the metering layer and the analytical target are not equivalent: cumulative subsystem measurements must first be converted into interval consumption through meter selection, differencing, treatment of physically implausible negative steps, aggregation and outlier control before they can function as an hourly cooling-demand series.

For benchmark-oriented modelling, however, predictive usefulness is not the only criterion for input selection. Qaisar and Zhao (2022) stress that the meaning of a baseline depends on the chosen independent variables and Chiosa et al. (2022) similarly show that anomaly detection depends on

both a robust reference model and an appropriate definition of the features used to identify deviation from normal behaviour. In this setting, variables drawn from the same signal family as the target may improve fit while also reducing interpretability and increasing circularity risk. This issue is especially relevant when target-adjacent cooling proxies are available: they may strengthen predictive accuracy numerically, yet weaken the independence between explanatory inputs and the reconstructed target that the model is intended to benchmark.

A closely related strand is limited-metering benchmarking and abnormal-consumption detection. Canale et al. (2024) define abnormal thermal energy consumption as deviation from a reference value and show that diagnosis remains possible even when metering infrastructure is incomplete, provided that benchmarking is performed systematically and contextual variables are incorporated appropriately. Their study is particularly useful for the present thesis because it frames limited instrumentation not as a reason to abandon analysis, but as a reason to design the analytical pipeline more carefully. It also reinforces the idea that reference-based interpretation is essential: abnormality cannot be read directly from raw consumption values, but emerges only through comparison with an appropriate benchmark.

This logic also connects naturally to baseline modelling and measurement and verification. Qaisar and Zhao (2022) describe the energy baseline model as a reference estimate used to evaluate performance under changing conditions, while Alrobaie and Krarti (2022) show that data-driven baseline models are increasingly used for M&V applications when historical operational data are available. In the present thesis, this bridge is important because target engineering is not an end in itself: it matters because the reconstructed cooling series is later used not only for forecasting, but also as the basis for expected-versus-actual operational benchmarking.

Taken together, the literature supports treating target engineering as a substantive part of the thesis contribution rather than as a hidden preprocessing detail. In many forecasting studies, the target is assumed to be directly available as hourly load, temperature, or total energy demand. In the present case, by contrast, the hourly cooling target must be identified, screened, reconstructed from fragmented cumulative subsystem measurements and cleaned before forecasting can even begin. The literature on preprocessing, sub-metering, limited metering and baseline-oriented modelling shows that this is not a peripheral inconvenience but a central methodological issue in real-world energy analytics.

1.4 Short-term cooling demand forecasting: model families, predictors and horizon effects

Short-term cooling-demand forecasting is typically approached through three broad modelling families: physics-based models, data-driven models and hybrid or grey-box models. For real operational datasets, data-driven approaches are often preferred because they can be trained directly on measured telemetry without requiring a fully specified physical description of the building or cooling system. Chen et al. (2022) and Khalil et al. (2022) both show that model suitability depends on the forecasting objective, the available system information, the temporal resolution of the data and the intended use of the model. In this context, a comparative modelling strategy is methodologically more appropriate than assuming in advance that either statistical models or deep-learning models should dominate.

The evaluation design is also a methodological choice rather than a neutral reporting detail. Mariano-Hernández et al. (2021) introduce a backtesting stage before final model selection, while Alrobaie and Krarti (2022) show that conclusions from data-driven energy models depend not only on the modelling algorithm but also on the adopted validation framework. For operational time series, this implies that chronological holdout testing alone may provide weaker evidence than rolling or expanding-window evaluation, which better reflects changing temporal conditions and repeated forecast deployment. The same logic also supports reporting more than one error metric. In applied energy forecasting, MAE and RMSE do not emphasise the same aspects of predictive performance, since RMSE penalises larger deviations more strongly, while MAE gives a more direct measure of average forecast error. As a result, comparative conclusions can differ depending on the chosen metric and both measures are useful when models are later interpreted operationally rather than only statistically.

Within this broader field, the most relevant literature for the present thesis is the cooling-load forecasting literature. Guo et al. (2014) model hourly cooling load using time-indexed autoregressive models with exogenous inputs and explicitly treat cooling demand as a function of both recent load history and external weather conditions. Their formulation is particularly important because it reflects the inertia of large cooling systems: demand does not respond instantaneously to changing external conditions and recent load values therefore contain essential predictive information. The same study also shows that weather variables such as outdoor temperature and relative humidity remain relevant, but that their influence is filtered through system memory and varies with forecast horizon. For a telecom cooling study based on operational telemetry, this provides a strong conceptual basis for treating cooling demand as simultaneously autoregressive and weather-sensitive rather than as a purely exogenous response.

A second important strand concerns nonlinear machine-learning and deep-learning methods. Fan et al. (2017) show that nonlinear approaches are effective for short-term building cooling-load prediction and that deep learning can improve performance, especially when used for feature extraction. At the same time, their results do not support the assumption that supervised deep models automatically dominate simpler alternatives. Earlier next-day work by Fan et al. (2014) also shows that ensemble approaches can outperform individual predictors, while tree-based and other nonlinear models remain among the strongest individual methods in building-energy forecasting. A similar pattern appears in technical-infrastructure transfer literature. Kula et al. (2025), working with a monitored data-centre cooling system, compare modern neural architectures with Random Forest, XGBoost and AutoARIMA and show that tree-based methods remain highly competitive even when newer neural approaches are included. Taken together, these studies suggest that in cooling-related operational datasets, nonlinear tree-based models deserve to be treated as first-class benchmarks rather than as secondary baselines behind recurrent neural networks.

Predictor design is equally important. Khalil et al. (2022) identify historical demand, exogenous variables such as meteorological information and contextual features as central inputs in data-driven building-energy forecasting. Cooling-specific studies refine this pattern further. Mariano-Hernández et al. (2021) combine historical, meteorological, calendar and past-series variables in a direct multi-step forecasting strategy, while Chalapathy et al. (2021) note that many buildings lack high-granularity contextual variables such as detailed occupancy information or local ambient measurements at matching resolution. Lin et al. (2022) make a similar point by showing that practical variable selection is often constrained by missingness and incomplete telemetry rather than by theoretical completeness. This is directly relevant to the present thesis, where dense indoor meteo sensing is unavailable, but a sparse internal temperature signal could still be reinterpreted as a slow contextual thermal-state proxy alongside external weather and operational variables. At the same time, sparse sensing introduces an interpretive boundary: a model may forecast expected cooling demand well without directly observing indoor thermal state. This issue becomes more important when some candidate predictors are closely related to the target-construction chain itself, because improved predictive accuracy in such cases may come at the cost of weaker independence between drivers and response.

Forecast horizon further changes the nature of the problem. Guo et al. (2014) use different ARX formulations for shorter and longer horizons, showing that very short-term cooling prediction depends more heavily on recent load history, while longer horizons require greater reliance on

broader temporal structure and exogenous inputs. Chalapathy et al. (2021) arrive at a similar conclusion in multi-step experiments and demonstrate that model rankings can change with horizon and with the quantity of available training data. The same literature also shows why naive persistence remains a serious benchmark rather than a trivial one. Chalapathy et al. (2021) retain a last-value baseline in multi-step experiments and Guo et al. (2014) show that short-horizon cooling demand is strongly shaped by recent load history. This supports the interpretation that strong naive performance at one hour ahead is a structural property of the problem, not a sign of weak modelling design. More generally, the literature suggests that very short-horizon cooling forecasting is typically persistence-dominated, whereas at longer horizons the relative contribution of exogenous structure and model form becomes more visible.

Table 1.1 summarises the main intelligent-control strategy families reviewed in the literature and positions the present thesis within that broader control landscape.

Strategy family	**Main idea and requirements**	**Relevance to the thesis**
Rule-based supervisory control	Predefined thresholds, schedules, or logic; requires only basic measurements and domain rules	Useful benchmark and conceptual baseline; common in practice (Maddalena et al., 2020; Stoffel et al., 2023)
Predictive control (MPC)	Uses a model, forecasts, constraints, and manipulated variables to optimize future control actions	Core intelligent-control literature anchor, but full implementation requires richer states and actuator-level data than available here (Killian and Kozek, 2016; Drgoňa et al., 2020; Xin et al., 2024)
Data-driven / approximate predictive control	Replaces or simplifies physical models through machine-learning models or learned MPC-like policies; lowers modelling burden but still needs structured control setup	Most relevant methodological bridge from forecasting to control; supports supervisory logic derived from predictive models (Yang et al., 2021; Klaučo et al., 2014; Smarra et al., 2018; Stoffel et al., 2024)
Model-free predictive control / reinforcement learning	Learns control behaviour directly from data or interaction without an explicit plant model	Reviewed for completeness, but less suitable here because it typically requires richer feedback, training environment, and safety assurance

		(Maddalena et al., 2020; Xin et al., 2024; Stoffel et al., 2023)
Cooling-specific predictive control in technical infrastructure	Applies predictive control directly to cooling supply, airflow, or thermal balancing under infrastructure-specific constraints	Closest transfer literature to telecom cooling; shows what full cooling-specific control looks like when zone states and control variables are available (Zhao et al., 2025)

Table 1.1 Intelligent control strategies reviewed for cooling systems and relevance to the thesis

As shown in Table 1.1, the control methodology adopted in this thesis is best classified as supervisory predictive control. This means that control decisions are made at a supervisory level using forecast outputs and contextual operating information, rather than through fixed schedules alone or through full online optimization. In the present study, supervisory predictive control is implemented in a simplified rule-based simulation form: the forecast model is used to identify candidate intervention periods, and bounded control actions are then triggered only when additional operational conditions are satisfied. The thesis therefore does not implement full MPC, data-driven MPC, or model-free control, but instead applies a simplified supervisory predictive control strategy that is consistent with the available telecom telemetry and with the scope of historical simulation.

Overall, the literature supports a horizon-sensitive interpretation of cooling-demand forecasting. At very short horizons, cooling demand is typically dominated by recent history, whereas at longer horizons the relative importance of exogenous structure and model form may increase. In the present thesis, the methodological focus is on short-horizon hourly forecasting, where persistence, lagged structure, and contextual refinement are most relevant to later benchmark-based interpretation and forecast-informed supervisory control design.

1.5 Predictive and intelligent control strategies for cooling systems

Intelligent control of HVAC and cooling systems is most commonly framed in the literature as a predictive-control problem. Killian and Kozek (2016) show that buildings are well suited to model predictive control because they involve slow dynamics, competing objectives and important future disturbances such as ambient temperature, solar radiation and occupancy. Drgoňa et al. (2020) extend this view by treating MPC not merely as a single algorithm, but as a full methodology involving model formulation, optimization, uncertainty handling, implementation architecture and standardized performance assessment. A more recent review by Xin et al. (2024) similarly

confirms that model-based predictive control remains the dominant mature branch of intelligent HVAC control, while model-free predictive control is expanding but remains less established in practical deployment. Together, these studies indicate that intelligent control in buildings is defined primarily by predictive, constraint-aware decision-making rather than simply by the use of artificial intelligence methods.

A second strand of the literature focuses on reducing the practical barriers of advanced control through data-driven methods. Maddalena et al. (2020) review rule-based control, reinforcement learning, MPC and learning MPC and note that industrial HVAC systems still rely heavily on PID regulators and rule-based supervisory schemes despite the growing academic interest in more rigorous approaches. They argue that data-driven and learning-based predictive control are attractive because they may reduce deployment time and modelling cost, but they also emphasize obstacles such as low-quality measurements, limited excitation and the continued need for safe and trusted supervisory behaviour. Kathirgamanathan et al. (2021) reach a similar conclusion from a broader data-driven predictive-control perspective, identifying feature selection, benchmarking, data quality and scalability as major unresolved methodological issues. This literature is directly relevant to real cooling infrastructure because it shows that the practical challenge is not only to formulate an optimal controller, but to do so under imperfect sensing, imperfect models and operational constraints.

The literature also shows that intelligent control should be evaluated comparatively rather than only against weak conventional baselines. Stoffel et al. (2023) address this explicitly by comparing six advanced control strategies under the same use case and with both quantitative and qualitative criteria: a well-designed rule-based controller, three MPC variants based on white-box, gray-box and black-box modelling, an approximate MPC and a reinforcement-learning controller. Their study is particularly important because all controllers are assessed under shared boundary conditions and against a credible rule-based benchmark rather than against a deliberately weak reference. Compared with that well-tuned rule-based controller, all advanced controllers reduced thermal discomfort and energy use, with the adaptive black-box MPC performing best in their case and achieving 8.4% energy savings. Stoffel et al. (2024) extend the practical perspective further by showing that higher prediction accuracy does not automatically yield the best controller in deployment: although artificial neural networks provided the highest prediction accuracy, the observed control-performance differences were small and linear regression models required less tuning effort and were recommended for practice. These studies are highly relevant because they

show that intelligent control must be judged not only by nominal optimality, but also by engineering effort, interpretability, data requirements and implementability.

A further important strand concerns simplified or approximate predictive control. Yang et al. (2021) demonstrate that an approximate MPC learned from measured operation data can retain most of the energy and comfort benefits of full MPC while being substantially faster computationally. Klaučo et al. (2014) approach the same problem from another direction by learning simple MPC-like feedback laws from closed-loop data and encoding the resulting policy as a regression-tree-based piecewise affine control law that can be implemented on simple hardware. The significance of these studies is methodological: they show that intelligent control need not always take the form of full online optimization. In some settings, the essential decision logic of predictive control can be approximated, learned, or expressed as a simpler supervisory policy that preserves much of the original control value while reducing implementation burden. This literature is especially important for the present thesis because it legitimizes a supervisory interpretation of intelligent control rather than restricting the term to full MPC deployment.

In data-limited supervisory control settings, an additional design step may involve operating-mode discovery from historical telemetry rather than direct plant-state modelling. Internal clustering criteria such as silhouette width and the Davies–Bouldin index are standard tools for assessing candidate partitions. Silhouette analysis evaluates how well each observation fits within its assigned cluster relative to neighbouring clusters, whereas the Davies–Bouldin index evaluates the trade-off between within-cluster compactness and between-cluster separation (Rousseeuw, 1987; Davies and Bouldin, 1979). At the same time, comparative studies of cluster validity indices show that index performance depends on data structure, cluster geometry and the clustering algorithm used, so no single internal criterion should be treated as universally decisive across applications (Arbelaitz *et al.*, 2013). For this reason, clustering-based mode design is strongest when numerical validation is combined with operational interpretability, allowing the resulting modes to function as conservative supervisory contexts rather than as hard physical state labels.

This bridge is particularly relevant when the forecasting layer is strongest in tree-based machine learning rather than in physics-based models. Drgoňa et al. (2020) note that random forest and regression-tree approaches have already been implemented successfully in optimal building control, although such models may also face limitations such as overfitting in some scenarios. Smarra et al. (2018), cited both in the broader MPC literature and in later comparative-control studies, provide a clear example of this direction by integrating random forests into a data-driven predictive-control framework for building energy optimisation. The relevance of this strand to the

present thesis is direct: if the best-performing forecasting family in the cooling-demand problem is lagged tree-based modelling, then a move toward forecast-informed supervisory control is methodologically more coherent than a sudden move to a high-fidelity white-box controller or reinforcement learning without the required state and actuator information.

Cooling-specific studies in technical infrastructure provide the closest transfer literature. Zhao et al. (2025) develop a cooling-supply-allocation MPC for a multi-zone data-center CRAC system that explicitly addresses thermal coupling, overcooling, undercooling and cooling-energy waste. Their controller combines a multi-zone temperature prediction model with dynamic optimization of supply-air temperature and airflow and the reported results show reduced inter-zone temperature imbalance together with an 8.73% reduction in total cooling supply relative to the baseline strategy while maintaining acceptable temperature limits. This study is important not because it can be reproduced directly in the present thesis, but because it shows what a fully specified intelligent cooling-control study looks like when manipulated variables, zone-temperature feedback and validated operating constraints are available. It therefore provides a useful upper benchmark for defining the scope of the current research.

Additional data-center cooling studies make this transfer bridge more concrete. Lazic et al. (2018) show that data-driven model-predictive control can be deployed in a large-scale commercial data center with limited prior knowledge, highlighting the practical value of receding-horizon optimization under real operational constraints. Mirhoseininejad et al. (2020) extend this direction by proposing a data-driven multi-setpoint predictive thermal control system that jointly manages cooling units and heterogeneous thermal requirements, making the problem structurally closer to infrastructure cooling than generic building HVAC control. Zhao et al. (2024) further show that data-center cooling control can be formulated at whole-system level in multi-chiller operation, where energy conservation and thermal stability must be balanced explicitly. Together, these studies strengthen the transfer argument from general building MPC to cooling-specific technical infrastructure, while also clarifying why the present thesis remains at the level of forecast-informed supervisory control simulation rather than claiming deployable closed-loop MPC.

Taken together, the reviewed literature supports a layered understanding of intelligent control for cooling systems. At the most developed end are full predictive-control approaches such as MPC, including white-box, gray-box, black-box and data-driven variants. A second layer includes approximate or learned predictive controllers that preserve much of MPC's logic while reducing computational or modelling burden. A third layer consists of supervisory control strategies informed by predictive models, extracted control rules, or forecast-based decision logic. The

present thesis is positioned closest to this third layer. It does not implement validated closed-loop MPC or reinforcement learning in a telecom central office, because the available telemetry does not provide actuator-level command variables, rich indoor thermal-state feedback, or a safe online control environment. Instead, the experiment implements forecast-informed intelligent supervisory control strategy simulation using machine-learning forecasting, bounded actions, explicit control states and K-means-based operating-mode discovery and compares these strategies against historical operation. In this sense, the thesis contributes a defensible intermediate control layer between pure forecasting and full closed-loop intelligent control.

1.6 Forecast-based benchmarking, reduction-potential interpretation and thesis positioning

Once predictive modelling and supervisory control logic have been established, the next methodological question is how their operational value should be evaluated. In building-energy research, this is typically addressed through baseline modelling and measurement-and-verification logic. Qaisar and Zhao (2022) describe the energy baseline as a reference estimate of how a building would be expected to consume energy under given conditions, while Alrobaie and Krarti (2022) show that predictive models become practically useful when they serve as defensible expected-demand baselines rather than as isolated forecasting tools. In the context of the present thesis, this means that both the forecast model and the simulated supervisory strategies must be interpreted relative to historical operation and expected demand, not only through prediction error but through their implications for benchmark-based cooling assessment.

This benchmarking perspective is particularly important when cooling demand is reconstructed from incomplete or fragmented metering and when the available data do not provide direct thermal-comfort or indoor-state confirmation. Canale et al. (2024) define abnormal thermal consumption as a significant deviation from a reference value and show that such diagnosis remains possible under limited metering when benchmarking is performed systematically. Chiosa et al. (2022) likewise argue that anomaly detection depends on an appropriate context-sensitive baseline and on careful feature selection, rather than on raw consumption inspection alone. For the present thesis, these studies support a restrained interpretation of positive residuals and control-trigger hours: they are best understood as model-relative evidence of possible excess cooling demand, not as direct proof of physical overcooling.

Weather normalization adds a further layer of caution to this interpretation. Walsen et al. (2025) show that weather-normalized comparison is essential when energy performance is interpreted over time, because unadjusted differences may reflect climatic composition as much as operational

change. Qaisar and Zhao (2022) similarly emphasize that the meaning of an energy baseline depends strongly on the selected independent variables, especially when weather-sensitive end uses are being evaluated. For cooling-energy applications, this has a direct implication: annual and warm-season conclusions may diverge if residuals are aggregated without sufficient weather and season context. A season-aware comparison therefore provides a stronger basis for discussing possible excess cooling and simulated reduction opportunities than a single annual residual balance alone.

A further issue concerns how reduction potential should be interpreted once forecast-based or supervisory control strategies have been simulated. The literature does not support translating all positive residuals or all controller activations directly into actionable savings. Alrobaie and Krarti (2022) note that M&V outcomes remain sensitive to modelling choices and application conditions, while Chiosa et al. (2022) show that anomalous behaviour should be prioritised in context rather than treated uniformly. Stoffel et al. (2023) strengthen this point from the control perspective by showing that advanced control strategies must be compared under common conditions and against credible baselines if their operational value is to be interpreted meaningfully. In the present thesis, this supports the use of bounded actions, persistence requirements, operating-mode filters and comparison against historical operation as a conservative way to evaluate forecast-informed supervisory control strategies rather than as a claim of directly achievable savings.

The specific research gap addressed by this thesis lies at the intersection of telecom cooling diagnostics, short-horizon cooling-demand forecasting, and forecast-informed supervisory control under fragmented operational telemetry. Existing telecom studies are strong in monitoring, KPI design and diagnosis, but provide less direct support for reconstructing a usable cooling-demand target from distributed cumulative meters, forecasting that target at short hourly horizons, and translating forecast outputs into benchmark-based supervisory control logic. Forecasting studies, meanwhile, often assume ready-made targets and stop at predictive accuracy, whereas advanced-control studies usually assume richer state feedback, manipulated variables and validated control environments than are available in the present telecom case. This thesis addresses that gap through an applied framework for a real telecom central office that combines target engineering, comparative rolling-origin forecasting, expected-versus-actual benchmarking, and simulation of forecast-informed supervisory control strategies on historical operation.

Finally, the literature supports the thesis at four linked levels. First, telecom central-office studies justify treating cooling as a benchmark-oriented operational problem rather than as generic HVAC background. Second, preprocessing, sub-metering and target-engineering literature supports the

reconstruction of an hourly cooling-demand target from fragmented cumulative measurements. Third, forecasting literature supports comparative time-series modelling under strong autoregressive structure, weather sensitivity and horizon-dependent behaviour. Fourth, predictive-control and benchmark-oriented evaluation literature show that model outputs can be used not only for prediction, but also for supervisory decision logic and cautious historical comparison. The specific contribution of the present thesis lies in combining these strands within one real telecom-site case under realistic sensing constraints.

2. RESEARCH DESIGN, DATA AND METHODS

2.1 Research design and analytical workflow

This thesis adopts an applied quantitative research design based on retrospective analysis of historical monitoring telemetry from a selected telecom central office. The study is empirical and observational. It analyses previously recorded operational data rather than data generated through controlled experiments or live intervention. This design is appropriate because the objective of the thesis is to develop predictive models for cooling energy use and to evaluate intelligent control strategies under real operational constraints, not to validate new physical equipment or conduct actuator-level control experiments. In this sense, the research addresses the prediction, benchmark-based assessment, and control-oriented analysis of cooling energy performance as an operational problem in a mission-critical technical environment.

The context of the study plays an essential role in the research design. Telecom central offices are continuously cooled environments in which thermal stability is required for the reliable operation of sensitive electronic equipment. At the same time, the available data are not structured as a ready-made analytical dataset. Instead, the research must begin from imperfect monitoring telemetry characterized by fragmented subsystem measurements, uneven completeness, and limited direct thermal-state information. For this reason, the study is designed as a single-site case study that prioritises methodological depth, traceability, and operational realism over broad but shallow cross-site comparison.

The research design is structured around three linked analytical questions. The first asks how real operational and environmental data from a telecom central office can be analysed and transformed into a defensible hourly cooling-demand target and analytical dataset. The second asks which modelling approach provides the most reliable short-term benchmark for expected cooling demand. The third asks to what extent forecast-informed intelligent control strategies, evaluated through historical simulation, can identify conservative opportunities to improve cooling energy use. These questions are not independent. Each later stage of the study depends on the validated outcome of the previous one, and the research design therefore follows a sequential analytical logic rather than a set of isolated experiments.

This structure is consistent with the research gap identified in the literature review. The reviewed telecom literature is comparatively strong in monitoring, KPI development, and diagnosis, but weaker in the full chain from target reconstruction to short-horizon cooling forecasting,

benchmark-based operational assessment, and supervisory control logic under sensing constraints. The contribution of the thesis is therefore not the proposal of a new control algorithm or a new forecasting architecture in isolation. Rather, it is the development of a methodologically integrated applied framework that combines target engineering, comparative forecasting, benchmark-based evaluation, and intelligent control simulation within one real telecom central office case.

WIP

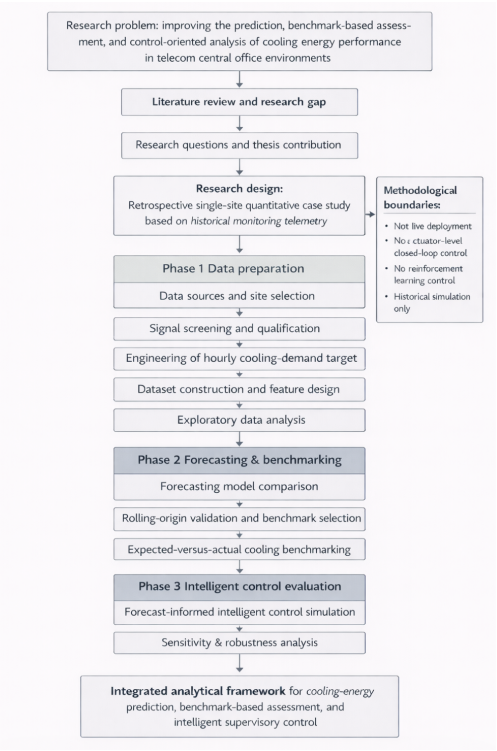

Figure 2.1 Research design and analytical workflow of the thesis

The overall analytical workflow of the thesis is summarised in Figure 2.1. The workflow begins with source review and selection of a suitable telecom site. It then proceeds to signal screening and qualification, engineering of an hourly cooling-demand target from fragmented cumulative subsystem telemetry, dataset construction and feature design, and exploratory data analysis. On that basis, short-term forecasting models are developed and compared under chronological and rolling-origin validation in order to select a defensible expected-demand benchmark. The selected

benchmark is then used for expected-versus-actual cooling assessment and for the simulation of forecast-informed intelligent control strategies. The final stage consists of sensitivity and robustness analysis of the supervisory design.

Although the workflow is primarily data-driven, it is not purely automatic. Engineering judgement enters at several points, particularly in site selection, signal qualification, target-component review, and interpretation of supervisory operating modes and control states. This is especially important because the cooling target is not directly available as a ready-made hourly variable. It must first be reconstructed from fragmented cumulative subsystem measurements before forecasting, benchmarking, or intelligent control evaluation can be justified. For this reason, preprocessing and target engineering are treated in this thesis as substantive parts of the research design rather than as hidden technical preparation.

Within this design, predictive modelling serves two functions. The first is comparative and methodological: to determine which model family best captures short-horizon cooling demand in the selected context. The second is operational: to generate an expected-cooling benchmark against which actual cooling behaviour can be interpreted. Forecasting accuracy is therefore not treated as an end in itself. It matters because the selected model becomes the analytical reference for later benchmark-based assessment and for the residual logic used in intelligent control simulation.

The intelligent control component of the thesis is implemented at the supervisory level. It does not represent live deployment, actuator-level industrial control, full model predictive control implementation, or reinforcement-learning control. Instead, the study evaluates forecast-informed intelligent control strategies through historical simulation. These strategies use forecast deviations, contextual variables, explicit control states, operating modes, and bounded reduction rules to test whether conservative intervention opportunities can be identified under real telemetry constraints. This scope is methodologically consistent with the available data and with the position of the thesis in the literature review, where supervisory control is treated as a defensible intermediate layer between pure forecasting and full closed-loop advanced control.

Overall, the research design can be described as a stage-gated analytical framework in which each phase produces the validated input for the next. The study moves from raw monitored data to operational insight without assuming that either the target variable or the control logic already exists in ready-made form. Its contribution lies in bridging site-level telemetry, engineered cooling-demand reconstruction, comparative short-term forecasting, expected-versus-actual

benchmarking, and forecast-informed intelligent control simulation within one coherent empirical study of cooling energy use in a telecom central office.

2.2 Study site and data sources

This section describes the database access context, the visible source tables, the basis for site selection, and the rationale for the selected study period.

2.2.1 Data access and extraction context

The empirical analysis was based on historical monitoring data obtained through read-only query access to a reporting database. Data extraction and inspection were performed using Python scripts. The first step was not full measurement loading, but an initial technical check of what was visible in the accessible reporting layer. A `SHOW TABLES` query confirmed that the visible tables were `datapoint`, `enmos_datapoint`, `measurement`, `measurement_all`, `measuring_point`, `mobnot`, `mobnot_reported_state`, `monitoring_site`, `monitoring_site_model`, and `swenex_file_data`.

After confirming the visible tables, a lightweight metadata inventory was created before any heavy time-series extraction was attempted. Column metadata from the visible tables were exported to `columns_metadata.csv`, and a separate helper file, `tables_relations.csv`, was used to keep track of practical links between selected sources, for example between `monitoring_site_model` and `monitoring_site`, `enmos_datapoint` and `monitoring_site`, and `swenex_file_data` and `monitoring_site`.

This access structure determined the order of the analytical workflow. Because the measurement layer was considerably heavier than the metadata layer, the study began with source discovery and metadata inspection rather than immediate bulk extraction of measurement records. In practical terms, `monitoring_site` and `monitoring_site_model` provided site-level context, `enmos_datapoint`, `datapoint`, and `measuring_point` described the available signals, `measurement` and `measurement_all` contained time-stamped values, and `swenex_file_data` provided supporting utility-related information used during initial screening.

This is important for the interpretation of the study data. The final analytical dataset was not available as one ready-made table. It had to be assembled progressively from several visible reporting sources, starting from table discovery, moving to metadata review, and only then proceeding to site selection, signal qualification, and later target construction.

2.2.2 Accessible source tables and their analytical roles

The accessible reporting layer was not a single modelling table, but a small relational source set consisting of ten visible tables: `monitoring_site`, `monitoring_site_model`, `swenex_file_data`, `enmos_datapoint`, `datapoint`, `measuring_point`, `measurement`, `measurement_all`, `mobnot`, and `mobnot_reported_state`. The initial discovery step confirmed these visible objects and a metadata export was created before large-scale extraction began.

From an analytical point of view, the visible source layer can be divided into three main groups. The first group is the site and support layer, represented mainly by `monitoring_site`, `monitoring_site_model`, and `swenex_file_data`. The second group is the signal metadata layer, represented by `enmos_datapoint`, `datapoint`, and `measuring_point`. The third group is the measurement layer, represented by `measurement` and `measurement_all`. Table 2.1 summarises these source layers and their roles in the study, while a full technical inventory of visible tables, selected fields, and practical inter-table relations is provided in Annex 2.

Table 2.1 Source layers and their analytical roles in the study

Source layer	**Main tables**	**Main role in the study**
Site and support layer	`monitoring_site`, `monitoring_site_model`, `swenex_file_data`	Site identification, site-model context, supporting utility information, and initial candidate-site screening
Signal metadata layer	`enmos_datapoint`, `datapoint`, `measuring_point`	Signal identification, metadata interpretation, cross-table mapping, and definition of the candidate signal universe
Measurement layer	`measurement`, `measurement_all`	Time-stamped signal values used for signal profiling, hourly aggregation, target construction, and dataset assembly
Visible but not used in the final workflow	`mobnot`, `mobnot_reported_state`	Present in the accessible reporting layer, but not used in the final analytical pipeline

The site and support layer provided the structural and contextual basis for site selection and source interpretation. In particular, `monitoring_site_model` added model-level context for the selected site, including versioned site-model attributes and a structured technical site description through the `doc` field. In the thesis, this document is treated as a technical semantic reference for interpreting the site infrastructure, especially cooling-related subsystems and other equipment groups. The signal metadata layer provided the descriptive basis for identifying and interpreting the available signal universe, while the measurement layer provided the time-stamped values used later for signal profiling, hourly aggregation, target engineering, and dataset construction. The two `mobnot` tables were visible in the reporting environment but remained outside the final analytical workflow.

In practice, the final thesis workflow depended mainly on the interaction between the site and support layer, the signal metadata layer, and the measurement layer. The analytical dataset was therefore assembled progressively from multiple visible reporting sources rather than extracted as one predefined modelling table.

2.2.3 Selected study site and site-selection basis

The study site was selected through a staged metadata-first screening pipeline designed to identify one telecom central office with sufficiently rich cooling-relevant telemetry before any heavy measurement extraction was attempted. This design was consistent with the overall stage-gated workflow of the thesis and was intended to reduce unnecessary loading of large time-series tables at an early stage. The visible reporting environment already contained the light tables required for this purpose, and the notebook explicitly defined the site-selection stage as an initial source-qualification step performed prior to measurement-layer extraction. The screening objective was to identify one site with strong cooling-related signal richness, energy-related metadata support, at least minimal temperature or context support, and utility-related support information for downstream analysis.

The first stage of the pipeline used `enmos_datapoint` as the primary source for site-level signal screening. For each site, textual metadata fields were normalized and combined from `name`, `description`, `source_class`, `status`, and `unit`. Four keyword groups were then applied - cooling, energy, temperature, and status. Based on these groups, boolean flags were assigned for cooling-related, energy-related, temperature-related, and status-related relevance. A metadata score was then calculated per signal using weighted keyword presence, where cooling and energy indicators received the highest weight, temperature indicators a slightly lower weight, and status

indicators a smaller supplementary weight. These signal-level scores were aggregated to site level to obtain `enmos_total_signals`, `enmos_cooling_signals`, `enmos_energy_signals`, `enmos_temp_signals`, `enmos_status_signals`, and `enmos_score`. This made the first ranking step fully reproducible from metadata alone.

In the second stage, site-level metadata from `monitoring_site` and `monitoring_site_model` were added to refine the ranking. Two additional quantities were derived at this stage: `has_site_model`, indicating whether the site had a linked record in `monitoring_site_model`, and `path_score`, derived from keyword matches in the site path metadata. These were combined with the aggregated ENMOS score to produce a first composite site score, `final_score = enmos_score + 3 × has_site_model + path_score`. The ranked list was then sorted by `final_score`, with the number of energy-, cooling-, temperature-, and total signals used as tie-breaking terms. At this stage, representative high-scoring signals from the top-ranked sites were also displayed and reviewed qualitatively to verify that the ranking was not dominated by irrelevant metadata noise.

In the third stage, the ranked sites were filtered to a reduced candidate set by requiring at least minimal energy and temperature support at metadata level. Operationally, this filter retained only sites with `enmos_energy_signals >= 1` and `enmos_temp_signals >= 1`. The purpose of this step was not to guarantee full modelling sufficiency, but to eliminate sites that were clearly unsuitable for downstream cooling analysis because they lacked even minimal energy or environmental-context evidence in the visible metadata layer.

In the fourth stage, the reduced candidate set was checked against `swenex_file_data` in order to add utility-related support information without touching the heavy measurement tables. For each candidate site, SWENEX coverage and support indicators were aggregated, including the number of rows, minimum and maximum available dates, total recorded consumption, total recorded cost, and counts of rows containing non-null cost and consumption values. From this, two additional variables were derived: `has_swenex_match`, indicating whether a site could be linked to SWENEX records, and `days_covered`, representing the length of the covered SWENEX period. These were combined with the earlier metadata score into a final composite experiment score: metadata relevance, presence of SWENEX support, date coverage, and cost/consumption support were all included, with upper clipping applied to prevent the utility terms from dominating the ranking. The final candidate ranking was then sorted by `experiment_score`, `final_score`, and the main SWENEX totals.

The final selected site was the top-ranked site under this combined procedure. Relative to the second-ranked candidate, it showed a substantially richer signal environment at metadata level, including 567 total ENMOS signals, 515 cooling-related signals, 548 energy-related signals, 4 temperature-related signals, and 13 status-related signals, whereas the second-ranked candidate showed only 90 total signals, 89 cooling-related signals, 78 energy-related signals, 1 temperature-related signal, and 10 status-related signals. The notebook therefore selected the top-ranked site as the experimental case site because it had the highest combined metadata and SWENEX-based screening score, the richest cooling and energy-related signal environment, and sufficient supporting information for downstream signal qualification and modelling. For confidentiality reasons, the site identifier is not reported in the thesis, throughout the text it is referred to simply as the selected telecom central office site.

This site-selection procedure is methodologically important because it makes the single-site design reproducible. The selected site was not chosen from prior domain familiarity or manual preference alone, but from an explicit screening pipeline that can be repeated on the same reporting environment using the same keyword groups, scoring logic, candidate filter, and utility-support enrichment steps. The result of this process was a technically justified site basis for the later stages of signal qualification, target engineering, dataset construction, forecasting analysis, and forecast-informed intelligent control simulation.

2.2.4 Study period and rationale for the selected time window

The accessible historical data covered more than one calendar year, but the empirical analysis was restricted to the 2025 calendar year. This was a deliberate study-design choice rather than a database limitation. The selected time window was intended to provide one recent, complete, and internally coherent annual period for downstream signal qualification, target engineering, forecasting, and benchmark-based analysis. At this stage of the workflow, the selection was justified at the source level rather than through the final engineered cooling target, because target identification and reconstruction were performed later in the chapter.

A month-level continuity check of the selected site measurement layer was therefore used to compare the available annual windows. As shown in Figure 2.2, the year 2025 provided the strongest full-year basis for downstream analysis. Monthly temporal continuity remained effectively complete throughout the year, with only one moderate reduction in August, where continuity reached 94%. By contrast, 2024 showed a visible disruption in spring, with continuity reduced to 74% in March and 20% in April, indicating that the annual record for that year was less

internally stable for the purposes of the planned analytical workflow. The 2026 year was not retained as the main study period because, although the available early months showed high continuity, the year was still partial at the time of extraction and therefore unsuitable as the principal annual basis for seasonal comparison, chronological forecasting validation, and full-year benchmark interpretation.

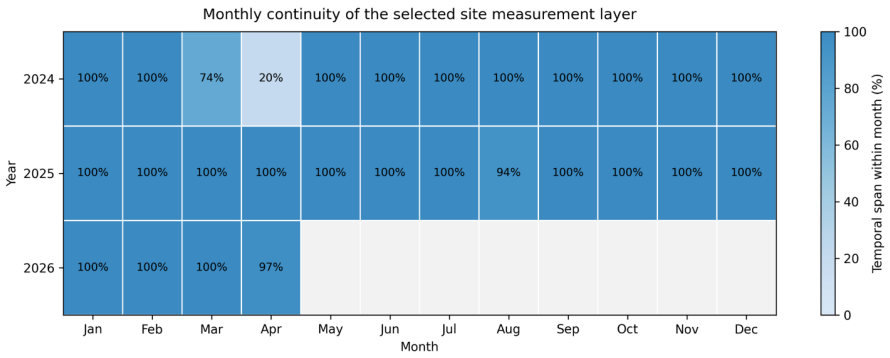

Figure 2.2 Monthly temporal continuity of the selected site measurement layer across available years

The use of 2025 as the study period was methodologically important for four reasons. First, the thesis reconstructs an hourly cooling-demand target from fragmented cumulative subsystem telemetry, and this requires a sufficiently continuous annual basis for differencing, aggregation, and plausibility checks. Second, one full annual cycle is needed to capture the seasonal structure of cooling demand within one internally consistent study window. Third, the forecasting stage relies on chronological evaluation and rolling-origin validation, both of which are more interpretable when built on one recent complete year. Fourth, the expected-versus-actual benchmark analysis and the later intelligent control simulation depend on comparable monthly and warm-season segments, which are more defensible when the study period covers the full annual cycle.

On this basis, the study period was restricted to the 2025 calendar year. Later stages of the methodology then used this selected annual window for signal qualification, engineering of the hourly cooling-demand target, construction of the modelling dataset, comparative forecasting, and forecast-informed intelligent control simulation.

2.2.5 Data categories and analytical roles

From the perspective of the analytical workflow, the accessible data can be divided into five categories: site-level metadata, technical site-model context, signal metadata, time-stamped measurement data, and supporting utility data. These categories were not equally important at every stage of the thesis. Instead, they entered the workflow sequentially, with different categories supporting source discovery, site selection, signal qualification, target engineering, forecasting, and later benchmark-based and supervisory analyses.

The first category is site-level metadata, represented mainly by `monitoring_site`. This category provided the basic site identifier and path-level context used to distinguish candidate sites and connect site-related records across the visible reporting layer. Its role was primarily structural rather than numerical: it supported source discovery and later linkage of site-specific information, but it did not itself provide the measurement values used in modelling. The second category is technical site-model context**,** represented mainly by `monitoring_site_model`. This category added model-related and contextual information for the selected site, including versioned site-model attributes and a structured technical site description through the `doc` field. In the thesis, this source is treated as a semantic reference that helped interpret the site's infrastructure structure, especially for cooling-related subsystems, utility groups, and other technical equipment families. Its geographic fields were also used internally to retrieve external weather data, although these values are not disclosed in the thesis for confidentiality reasons.

The third category is signal metadata, represented by `enmos_datapoint`, `datapoint`, and `measuring_point`. This category provided the descriptive layer required to understand what kinds of signals were available, how they were labelled, and how they could be connected to the measurement layer. `enmos_datapoint` was particularly important during candidate-site screening because it enabled keyword-based identification of cooling-, energy-, and temperature-related signal families at metadata level. `datapoint` then became the main signal registry used for the downstream experiment, while `measuring_point` added further structural context for how these datapoints were organised in the reporting environment. This category was therefore central to the transition from site-level source discovery to signal-level analytical qualification.

The fourth category is time-stamped measurement data, represented by `measurement` and `measurement_all`. This is the core empirical category of the thesis. `measurement` is the main normalized time-series source used for signal profiling, hourly aggregation, target engineering, and final dataset construction. `measurement_all` is a denormalized measurement view that

duplicates selected signal and measuring-point metadata together with time-stamped values. In the workflow, it was used mainly for source-consistency checks and exploratory lookup of signals that were visible in metadata but not directly mapped in the final datapoint-based signal set. The distinction between these two measurement sources became methodologically important because the thesis ultimately remained anchored on the datapoint-mapped signal universe rather than on ENMOS-only names.

The fifth category is supporting utility data, represented mainly by `swenex_file_data`. This category was used in the early screening stages to supplement metadata-based candidate-site ranking with utility-support information such as coverage period, consumption magnitude, and cost-related fields. Its role was therefore selective rather than central: it helped identify a suitable site for the experiment, but it was not used directly in target construction or forecasting. Similarly, the visible `mobnot` and `mobnot_reported_state` tables were part of the accessible reporting environment but remained outside the final analytical pipeline.

Taken together, these categories show that the thesis did not begin from a single ready-made modelling dataset. Instead, the empirical basis had to be assembled progressively from multiple source types with different analytical roles. This staged use of the data is consistent with the broader methodological position of the thesis: the final analytical framework combines source discovery, signal qualification, target engineering, forecasting, benchmark-based evaluation, and intelligent control simulation under real telemetry constraints rather than assuming that either the analytical target or the final modelling dataset already exists in prepared form.

2.3 Signal screening, cross-table mapping, and variable-selection logic

Commented [AZ4]: Figure Signal-screening and qualification pipeline for the selected site???

After establishing the database access context and the visible source structure, the next methodological step was to determine which signals from the selected site could actually support downstream analysis. The visible metadata layer contained more candidate signals than could be assumed usable for target engineering or forecasting, and the central task of this stage was therefore to move from nominal signal availability to a screened analytical signal universe. In the thesis, signal screening is understood as a staged qualification process in which metadata correspondence, measurement-layer availability, temporal continuity, and practical analytical relevance are evaluated before any final target or predictor set is defined. This step was necessary because the source environment contained multiple partially overlapping descriptions of the site's signals, but only a subset could be confirmed as stable time-series inputs for later modelling stages.

The signal-screening workflow began with cross-table mapping between `enmos_datapoint` and `datapoint` for the selected site. At this stage, 567 ENMOS metadata rows and 223 datapoint rows were loaded from the reporting layer for the site-specific signal environment. The key purpose of this comparison was to determine whether site-level ENMOS metadata and datapoint-level metadata described the same practical signal universe or only partially overlapping subsets. A direct mapping was then performed between `datapoint.dp_id` and ENMOS `name`. This showed that all 223 datapoint rows could be matched to ENMOS-side names, whereas 344 ENMOS rows remained unmatched from the datapoint perspective. The result was methodologically important. It showed that the datapoint layer formed a complete and internally consistent subset of the ENMOS-described site environment, while ENMOS also contained additional signals that could not automatically be treated as fully usable measurement-layer inputs.

The next step was measurement-layer profiling of the datapoint-mapped candidate pool. At this stage, the 223 matched datapoint IDs were tested against the `measurement` and `measurement_all` tables in order to determine practical signal availability, temporal range, and basic value behaviour. An initial profiling window was used first, after which the retained datapoint-mapped signals were later aggregated across the full annual study period. This profiling showed that not all metadata-visible datapoints produced a stable and usable measurement series, and the practically available datapoint-based exploration pool narrowed to 111 signals with measurement-layer support. A further consistency comparison between `measurement` and `measurement_all` for this retained pool showed complete overlap at the datapoint level for the profiled signals, which confirmed that the retained datapoint-mapped signal set behaved consistently across the two measurement sources.

A separate ENMOS-only availability probe was then conducted for the 344 ENMOS-side metadata rows that had no datapoint match. Because these signals lacked `datapoint_id`, they could not be queried through the normalized `measurement` table and were therefore tested only through a string-based lookup in `measurement_all`. This probe returned no confirmed matches under the tested lookup strategy. The result was interpreted conservatively: it did not prove that the ENMOS-only rows were meaningless, but it did show that they could not be established as directly retrievable time-series signals in the same way as the datapoint-mapped set. For this reason, the downstream analytical workflow remained anchored on the datapoint-mapped signal universe rather than on ENMOS-only metadata names. This decision was important for traceability, because later target engineering and dataset construction required signals that were both semantically interpretable and technically retrievable at the measurement layer.

After the cross-table and measurement-layer checks, the retained datapoint-based signals were transformed into a compact site-level signal inventory. This inventory combined measurement profiling with datapoint metadata and provided a working reference for signal identity, units, coverage, and variability. The same retained signal set was then aggregated to hourly resolution across the full selected study period, producing a multivariate hourly signal matrix for the selected site. The hourly summary confirmed that the retained datapoint-based signal set showed broad annual availability after hourly aggregation, and the resulting site representation formed the empirical basis for the next stages of the thesis - target-component review, target engineering, and predictor design. In this sense, the output of Section 2.3 was not yet a final feature set, but a qualified analytical signal universe from which later target and predictor selections could be made in a controlled and technically transparent way.

The logic of variable selection in this section was therefore deliberately sequential. Signals were not retained merely because they appeared in metadata, nor merely because they existed in one measurement source. Instead, a signal had to satisfy a minimum chain of conditions. It had to belong to the selected site, be interpretable through the metadata layer, be retrievable in the measurement layer, and contribute to a sufficiently continuous hourly representation of the site's monitored operation. This staged screening logic reduced the risk of basing later target engineering and forecasting on nominally visible but practically unstable or weakly supported signals. It also ensured that the final modelling workflow was grounded in a technically qualified signal environment rather than in the full but noisier raw metadata universe.

2.4 Engineering of the hourly cooling-demand target

A central methodological challenge of the thesis was that cooling demand was not available in the source environment as a directly usable hourly variable. Instead, the selected site contained a broader telemetry ecosystem composed of subsystem measurements, cumulative energy signals, contextual variables, and partially overlapping metadata descriptions. For the later forecasting, benchmarking, and intelligent control stages, a defensible hourly cooling-demand target therefore first had to be reconstructed from this fragmented signal environment. In the present thesis, target engineering is understood as the full process of identifying plausible cooling-related cumulative subsystem signals, testing their annual continuity and interpretability, converting cumulative measurements into interval consumption, handling physically implausible negative steps, aggregating the retained hourly increments, and reviewing the resulting series before it is used in downstream modelling.

This approach is consistent with the broader methodological position of the thesis. The literature reviewed in Chapter 1 shows that in real monitored infrastructure, operational data are rarely analysis-ready and that sub-item energy behaviour often has to be inferred from distributed measurements rather than from one ready-made metered variable. In that sense, the metering layer and the analytical target are not equivalent. For the present study, the cooling-demand target had to be engineered explicitly before forecasting could be treated as meaningful and before expected-versus-actual benchmarking or forecast-informed intelligent control simulation could be justified.

2.4.1 Selection of cumulative subsystem signals

The first step in target engineering was to identify which signals from the retained hourly signal universe could plausibly function as components of a site-level cooling-demand target. This selection was not performed mechanically from all cooling-labelled signals. Instead, it followed a staged review process that combined cooling-family metadata semantics, hourly usability after aggregation, non-trivial variability, strong co-movement with other plausible cooling-family signals, and interpretability within the technical site context. The objective at this stage was not yet to define the final target, but to construct a technically plausible candidate pool of cumulative cooling-related signals for closer comparison.

The initial candidate pool was intentionally broad. It included clearly labelled cooling-family cumulative energy signals as well as a smaller number of strongly linked infrastructure-adjacent signals retained temporarily for comparison. Table 2.2 summarises the shortlisted candidate signals and their technical role in the site cooling environment.

Table 2.2 Shortlisted cumulative cooling target candidates and their system role

Datapoint ID	**DP ID**	**Short description**	**System role**
120	V_K78_RZ	COOL Abwärmenutzung EWZ	Core cooling distribution / heat-recovery link
230	V_K81_RZ	COOL Ringleitung 1	Core cooling distribution
46	V_K82_RZ	COOL Ringleitung 2	Core cooling distribution
49	V_K76_RZ	COOL Kaltwasserperipherie Haupt	Core cooling distribution / periphery core
78	V_K74_RZ	COOL Steigstrang RK	Core cooling distribution
60	V_K71_RZ	COOL KM A Verdampfer	Production-side subsystem

221	V_K84_RZ	COOL Leitung B3 U6	Branch-level distribution
149	V_K83_RZ	COOL Leitung B1 U6	Branch-level distribution
232	V_K77_RZ	COOL Kaltwasserperipherie vor Abwärme	Peripheral / pre-heat-recovery branch
194	V_K75_RZ	COOL FC Umformer	Free-cooling / conversion-related subsystem
1	V_E86_RZ	COOL Ringleitung 1	Infrastructure-adjacent signal
201	V_E08_RZ	E08 Wirkenergie Trafo T9 U200	Infrastructure proxy
159	V_E06_RZ	E06 Wirkenergie Trafo T6 U200	Infrastructure proxy
91	V_E06B_RZ	E06B Blindenergie Trafo T6 U200	Infrastructure proxy

The monthly average profiles of the shortlisted candidates are shown in Figure 2.3. This figure should be interpreted as supporting visual evidence rather than as the sole selection rule. Its purpose is to compare the annual behaviour of the candidate signals and to distinguish the most stable and coherent cooling-family subgroup from candidates that, although cooling-related, display more peripheral, branch-like, or subsystem-specific behaviour.

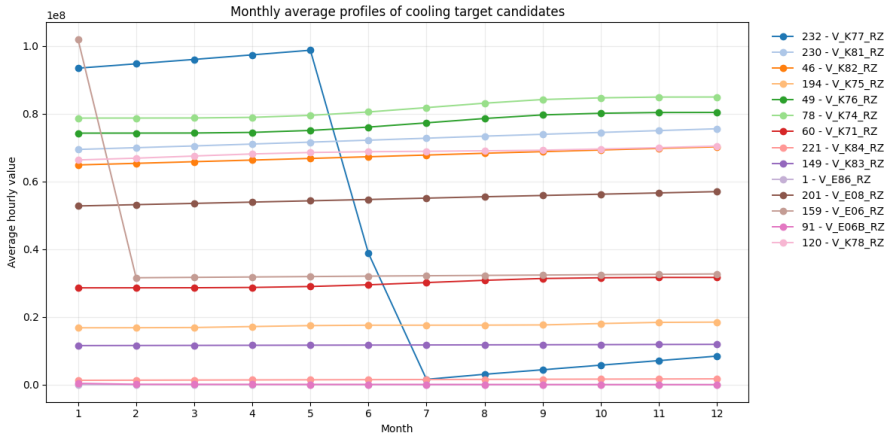

Figure 2.3 Monthly average profiles of cooling target candidates

Figure 2.3 shows that several shortlisted candidates exhibit broadly stable annual behaviour, but they do not all represent the same technical role in the cooling system. Signals such as `V_K71_RZ` and `V_K84_RZ` remain cooling-related and structurally plausible, yet they correspond to narrower production-side or branch-level subsystem roles rather than to the core distributed cooling

structure that the target was intended to represent. Similarly, signals such as `V_E08_RZ`, `V_E06_RZ`, and `V_E06B_RZ` showed strong linkage to the cooling environment but were interpreted more appropriately as infrastructure-adjacent or proxy signals than as components of the core cooling target.

On this basis, the final target components were restricted to datapoint IDs **120, 230, 46, 49, and 78**, corresponding to **V_K78_RZ**, **V_K81_RZ**, **V_K82_RZ**, **V_K76_RZ**, and **V_K74_RZ**. These signals were preferred because they jointly satisfied the full selection logic: they had clear cooling-family semantics, remained technically usable after hourly aggregation, showed non-trivial variability, formed a coherent co-moving signal group, and, most importantly, represented the most stable and interpretable core cooling-distribution subset among the shortlisted candidates. Candidates with narrower subsystem meaning, stronger branch-level interpretation, structural instability, or proxy-like infrastructure role were not retained in the final target definition.

At the end of this stage, the thesis therefore had a justified set of cumulative subsystem signals from which the hourly cooling-demand target could be constructed. The next step was to transform these retained cumulative measurements into hourly interval consumption through alignment, differencing, and treatment of implausible negative steps.

2.4.2 Hourly alignment and differencing

After the final cumulative target components had been selected, the next step was to transform them into a common hourly representation suitable for interval-based target construction. The retained signals were already available in the site-level hourly signal matrix produced in the previous stage of the workflow. For target engineering, the five selected cumulative components - datapoint IDs 120, 230, 46, 49, and 78 - were extracted from this hourly matrix and aligned on a shared hourly timestamp index. This ensured that all retained components were evaluated on the same temporal basis before any interval transformation was applied.

This alignment step was methodologically important because the retained signals represented cumulative subsystem measurements rather than direct hourly cooling demand. In cumulative form, the values describe running totals and therefore cannot be used directly as the target for forecasting or benchmark-based analysis. The objective of this stage was to convert the aligned cumulative series into hourly increments that more closely reflect interval cooling activity. Consistent with this logic, the retained hourly component matrix was first ordered chronologically

and then differenced along the time axis. In practical terms, the first-order difference of each cumulative series was used to estimate the change from one hourly timestamp to the next.

The differencing step changed the interpretation of the retained signals from cumulative subsystem totals to hourly interval contributions. This was a necessary transformation because the downstream analytical target of the thesis is an hourly cooling-demand variable rather than a stock-like counter. At the same time, differencing cumulative telemetry introduces known practical issues. In real metering environments, negative steps may occur because of counter resets, irregularities in data transmission, or other non-physical artefacts rather than true negative cooling consumption. For this reason, the differencing stage was treated as a distinct methodological step, while the treatment of negative or otherwise implausible steps was handled separately in the next subsection.

At the end of this stage, the thesis therefore had an aligned set of hourly interval series derived from the retained cumulative cooling components, but not yet the final target. The next task was to review these differenced values for negative and implausible behaviour before aggregation into a single hourly cooling-demand variable.

2.4.3 Treatment of negative or implausible steps
2.4.4 Aggregation into final target y

2.5 Dataset construction and feature design

- 2.5.1 Main dataset
- 2.5.2 Compare dataset
- 2.5.3 Weather and calendar variables
- 2.5.4 Operational variables
- 2.5.5 Internal temperature proxy

2.6 Exploratory data analysis
2.7 Forecasting methodology and validation design

- 2.7.1 Benchmark models
- 2.7.2 Machine-learning models
- 2.7.3 Lag structure
- 2.7.4 Rolling-origin validation
- 2.7.5 Evaluation metrics

2.8 Benchmarking and supervisory-control methodology

2.8.1 Expected-versus-actual cooling logic
2.8.2 Residual interpretation and reduction funnel
2.8.3 Supervisory strategy family B0–S3
2.8.4 Risk states and operating modes
2.8.5 Sensitivity-analysis design

3. FORECASTING RESULTS

3.1 Final dataset and target characteristics
3.2 Initial benchmark and model-family comparison
3.3 Rolling-origin validation and final model selection
3.4 Final one-hour-ahead forecast performance
3.5 Feature importance and ablation analysis
3.6 Compare-dataset findings
3.7 Six-hour-ahead sensitivity check

4. BENCHMARKING AND SUPERVISORY CONTROL SIMULATION RESULTS

4.1 Expected versus actual cooling benchmark
4.2 Full-year and warm-season interpretation
4.3 Reduction-potential funnel
4.4 Final supervisory strategy family

4.4.1 B0 simple threshold
4.4.2 S1 forecast-only
4.4.3 S2 load-aware
4.4.4 S3 load-and-mode-aware

4.5 Comparative strategy results
4.6 Risk-state, operating-mode and monthly behavior analysis
4.7 Sensitivity and robustness analysis

4.7.1 Proxy sensitivity
4.7.2 Cluster-number sensitivity
4.7.3 Threshold / persistence / cap sensitivity

4.8 Under-expected cooling diagnostic extension

5. DISCUSSION

5.1 Answer to RQ1: target engineering under fragmented telemetry
5.2 Answer to RQ2: short-horizon cooling forecasting
5.3 Answer to RQ3: forecast-informed supervisory control simulation
5.4 Methodological contribution of the thesis
5.5 Practical implications for telecom cooling analytics
5.6 Limitations and future work

CONCLUSIONS

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